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1 Confidence Intervals

1. Define i.i.d. variables $A_k \sim \text{Bern}(p)$ where $k \in \{1, \dots, n\}$. Assume we can declare that $\Pr \left[\left| \frac{1}{n} \sum_{k=1}^n A_k - p \right| \geq 0.25 \right] = 0.01$.

(a) Please give a 99% confidence interval for p given A_k .

Solution: Notice that the mean is $\frac{1}{n} \sum_{k=1}^n A_k$, and our confidence interval should cover the parts that are within 0.25 of the mean, leading to an interval of the form:

$$\left(\frac{1}{n} \sum_{k=1}^n A_k - 0.25, \frac{1}{n} \sum_{k=1}^n A_k + 0.25 \right)$$

(b) We know that the variables $X_1, \dots, X_n$, are i.i.d. random variables and have variance σ^2 . We also have the observation that $A_n = \frac{X_1 + \dots + X_n}{n}$. We want to estimate the mean, μ , of each X_i .

Prove that we have 95% confidence that μ lies in the interval $\left[A_n - 4.5 \frac{\sigma}{\sqrt{n}}, A_n + 4.5 \frac{\sigma}{\sqrt{n}} \right]$

That is, $\Pr \left[A_n - 4.5 \frac{\sigma}{\sqrt{n}} \leq \mu \leq A_n + 4.5 \frac{\sigma}{\sqrt{n}} \right] \geq 95\%$

Solution: To do this, we use Chebyshev's. Because $\mathbb{E}[A_n] = \mu$ (since A_n is the average of the X_i 's), we bound the probability that $|A_n - \mu|$ is more than the interval size at 5%:

$$\Pr \left[|A_n - \mu| \geq 4.5 \frac{\sigma}{\sqrt{n}} \right] \leq \frac{\text{Var}(A_n)}{\left(4.5 \frac{\sigma}{\sqrt{n}} \right)^2} \approx \frac{\frac{\sigma^2}{n}}{\frac{20\sigma^2}{n}} = \frac{1}{20} = 5\%$$

.

Thus, the probability that μ is in the interval is 95%.

(c) Give the 99% confidence interval for μ .

Solution: Solution is similar to that of the 95% confidence interval. Our 99% confidence interval is $\left[A_n - 10 \frac{\sigma}{\sqrt{n}}, A_n + 10 \frac{\sigma}{\sqrt{n}} \right]$ since

$$\Pr \left[|A_n - \mu| \geq 10 \frac{\sigma}{\sqrt{n}} \right] \leq \frac{\text{Var}(A_n)}{\left(10 \frac{\sigma}{\sqrt{n}} \right)^2} \approx \frac{\frac{\sigma^2}{n}}{\frac{100\sigma^2}{n}} = \frac{1}{100} = 1\%$$

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2. We have a die with 6 faces labeled 1, 2, 3, 4, 5, 6.

(a) Develop a 99% confidence interval for the sum of n samples.

Solution: Consider the 99% confidence interval for the average of n rolls of a standard die, $X = \frac{1}{n} \sum_{i=1}^n D_i$.

We can bound this using Chebyshev's inequality:

- (b) Now, suppose the die's face values are just 6 consecutive integers $k + 1, k + 2, \dots, k + 6$, but we do not know k . For example, if $k = 6$, the die faces would take on the values 7, 8, 9, 10, 11, 12. If we observe that the average of the n samples is 15.5, develop a 99% confidence interval for the value of k .

Solution: We can now say that the confidence interval for k is given by 15.5– confidence interval for X .

2 Conditional Expectation

1. Looping Ropes

Frobenius has n ropes in his backyard, which he likes a lot. But when he goes to work everyday, they grow a mind of their own and begin behaving weirdly. At every timestep t , two ends of a rope(s) are uniformly chosen at random and knotted together. If the two ends are from the same rope, they form a loop. If the two ends are from different ropes, they join together to form a new rope. By the time Frobenius comes home, this process has completed (meaning no more loose ends are left). How many loops can Frobenius expect to see? **Bonus:** Does this converge as $n \rightarrow \infty$?

Solution: let X_n be the number of loops when there are n ropes left. Then, at the next timestep, we can either pick two ends from the same rope or two ends from different ropes.

If we pick two ends from the same rope, we form one more loop and there are $n - 1$ ropes left. If we pick two ends from different ropes, we don't change the number of loops, and still the number of ropes decreases by 1 (since we combine two ropes into one). Let I be the event that we pick two ends of the same rope. Now, we can use conditional expectation to represent the number of loops we expect to see.

$$\mathbb{E}[X_n] = \mathbb{E}[X_n|I]\mathbb{P}[I] + \mathbb{E}[X_n|\bar{I}]\mathbb{P}[\bar{I}] \quad (1)$$

$$= (\mathbb{E}[X_{n-1}] + 1)\mathbb{P}[I] + \mathbb{E}[X_{n-1}]\mathbb{P}[\bar{I}] \quad (2)$$

$$= \mathbb{E}[X_{n-1}] + \mathbb{P}[I] \quad (3)$$

If there are n ropes, there are $2n$ ends and thus $\frac{1}{2n-1}$ probability of pick two ends on the same rope (after we pick the first, there are $2n - 1$ ends left). Thus $\mathbb{P}[I] = \frac{1}{2n-1}$. This gives us the recursive relation,

$$\mathbb{E}[X_n] = \mathbb{E}[X_{n-1}] + \frac{1}{2n-1} \quad (4)$$

with the base case $\mathbb{E}[X_1] = 1$ (because if there is one rope, both ends must on the same rope and thus form a loop). Fortunately, this is just the recursive relation for an arithmetic sequence, hence:

$$\mathbb{E}[X_n] = \sum_{i=1}^n \frac{1}{2i-1} = 1 + \frac{1}{3} + \frac{1}{5} + \dots \quad (5)$$

This can be represented as a partial sum of the difference of two harmonic series:

$$\sum_{i=1}^n \frac{1}{2i-1} = \sum_{i=1}^{2n} \frac{1}{i} - \frac{1}{2} \sum_{i=1}^n \frac{1}{i} = H_{2n} - \frac{1}{2}H_n \quad (6)$$

Since $H_{2n} > H_n$, we have that $H_{2n} - \frac{1}{2}H_n > \frac{1}{2}H_n$. Therefore,

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_n] \geq \lim_{n \rightarrow \infty} \frac{1}{2}H_n \quad (7)$$

Since the harmonic series doesn't converge neither does the number of loops.

2. If there are no distractions, it takes me 30 minutes to walk to the store. However, if I pass someone with a cat, I stop and pet the cat and chat to the owner. The number Y of cats I pass is a Poisson random variable with mean 2. Each time I stop, the number of minutes I spend petting the cat and chatting is an exponential random variable with PDF:

$$f_X(x) = 0.5e^{-0.5x}$$

- (a) If I see a single cat, what is the expectation and variance of the time spent petting the cat and chatting to its owner?

Solution: X is exponential, and so $E[X] = \int_0^\infty 0.5xe^{-0.5x}dx = 2$, $E[X^2] = \int_0^\infty 0.5x^2e^{-0.5x}dx = 8$, $\text{Var}(X) = E[X^2] - E[X]^2 = 4$

- (b) What is the conditional expectation $E[X | Y]$ of the total time spent petting cats and chatting to their owners, as a function of Y ?

Solution: $E[X | Y = y] = 2y$, (expectation of sum), so $E[X | Y] = 2Y$

- (c) Using the smoothing law (or law of iterated expectation, or law of total expectation), calculate $E[X]$.

Solution: $E[X] = E[E[X | Y]] = 2E[Y] = 2 \times 2 = 4$

3. Rolling Chopsticks

The content mentors were trying to eat noodles in a new way. Rather than eating noodles by chopsticks directly, they tried eating noodles by rolling one noodle on the chopstick and eat it. This is seemingly a hard way to eat noodles so the probability they successfully eat a noodle on each attempt is p .

- (a) Suppose they start attempting to eat a noodle, and eat the noodle on the attempt X . What is the distribution of X ? What is the distribution of unsuccessful attempts to eat that noodle, X' in terms of X ?

Solution: We notice that X is modeling how long it takes until the first successful attempt hence $X \sim \text{Geometric}(p)$. Now if $X = k$ we know the last one is the successful attempt so there are $k - 1$ unsuccessful attempts. That gives us $X' = X - 1$. So the distribution of unsuccessful attempts to eat that noodle $X' \sim X - 1$. That is $\mathbb{P}[X' = k] = \mathbb{P}[X = k + 1]$

- (b) Let Y be the unsuccessful attempts that they will make trying to eat 2 noodles. What is the distribution of Y ?

Solution: Now we know the distribution of the unsuccessful attempts to eat 1 noodle is X' . We need the distribution of the unsuccessful attempts that they will make trying to eat 2 noodles. So we could model $Y = X'_1 + X'_2$, where X'_1 and X'_2 are identical and independent random variables X' . Notice we know the distribution of X and so we need to substitute $X' = X - 1 \implies X = X' + 1$. Now for the distribution of Y , if $Y = k$ we observe that if we make w unsuccessful attempts before 1st successful attempt, so $X_1 = w + 1$, and then we will make $k - w$ unsuccessful attempts before the 2nd successful event, so $X_2 = k - w + 1$,

where w ranges from 0 to k .

$$\begin{aligned}
 \mathbb{P}[Y = k] &= \mathbb{P}[X'_1 + X'_2 = k] \\
 &= \sum_{w=0}^k \mathbb{P}[X'_1 = w \cap X'_2 = k - w] \\
 &= \sum_{w=0}^k \mathbb{P}[X_1 = w + 1 \cap X_2 = k - w + 1] \\
 &= \sum_{w=0}^k (1 - p)^w p \cdot (1 - p)^{k-w} p \\
 &= (1 - p)^k p^2 \cdot \sum_{w=0}^k 1 \\
 &= (k + 1)(1 - p)^k p^2
 \end{aligned}$$

- (c) Content with their distribution Y and eating 2 noodles, the content mentors want to find the distribution Z for the total unsuccessful attempts of eating the whole bowl of R noodles. They were planning to proceed as part b) but then Aekus, a random variable distribution enthusiast, suggested to use $P(Z = k) = \binom{r+k-1}{k} (1 - p)^k p^r$ where $r = R$.

The distribution Z is defined by 2 parameters: 1) the number of successful attempts r and 2) the probability of a successful attempt a so we will write Z as $Z(r, p)$.

Show that Z is the sum of independent X' random variables by using induction on r where $r = 1$ is the base case, and the content mentors can use Z as their distribution.

(Hint: Remember the "Hockey stick" identity $\sum_{i=0}^{k-1} \binom{n+i}{i} = \binom{n+k}{k-1}$)

Solution: For intuition let's try to put small values for r .

$$\begin{aligned}
 \mathbb{P}[Z(1, p) = k] &= \binom{1 + k - 1}{k} (1 - p)^k p^1 \\
 &= (1 - p)^k p^1 \\
 &= \mathbb{P}(X = k + 1) = \mathbb{P}(X' = k)
 \end{aligned}$$

When $r = 1$ we get that $Z(1, p) \sim X'$ and when $r = 2$ we get $Z(2, p) \sim Y$. Now this helps us to see that we could use our old friend induction to prove this.

Base Case: The base case is when $r = 1$ and is true as we saw above.

Hypothesis: Assume that $Z(r, p)$ is the sum of r X'_i , where $X'_i = X_i - 1$ for $X \sim \text{Geometric}(p)$ for $i = 1, \dots, n$ are IID.

Step : Now let $W = Z + X'$, if we show that W is the Z distribution with $r_W = r + 1$ we are done.

$$\begin{aligned}
 \mathbb{P}[W = w] &= \mathbb{P}[X' + Z = w] \\
 &= \sum_{k=0}^w \mathbb{P}[(Z = k) \cap (X' = w - k)] \\
 &= \sum_{k=0}^w \mathbb{P}[(Z = k) \cap (X = w - k + 1)] : \text{Using the substitution } X' = X - 1 \\
 &= \sum_{k=0}^w \binom{r + k - 1}{k} p^r (1 - p)^k \cdot p(1 - p)^{w-k} \\
 &= \sum_{k=0}^w \binom{r + k - 1}{k} p^{r+1} (1 - p)^w \\
 &= p^{r+1} (1 - p)^w \sum_{k=0}^w \binom{r + k - 1}{k} : \text{Using "Hockey Stick" with } n = r - 1, k = w + 1. \\
 &= \binom{r + w}{w} p^{r+1} (1 - p)^w,
 \end{aligned}$$

This shows that W is the Z distribution with $r_W = r + 1$ and probability of success as p , $Z(r + 1, p)$ and it is now evident that if $X'_i = X_i - 1$ where $X \sim \text{Geometric}(p)$ for $i = 1, 2, \dots, n$ are IID, then $\sum_i^r X'_i = Z(r, p)$
The distribution Z has a name, it's called the Negative Binomial Distribution, just a fun fact.

- (d) What is the expected value of total unsuccessful attempts of eating the whole bowl of R noodles, the random variable Z ?

Solution: From part c) we proved that Z is the sum of r independent X' random variables. Then

$$\begin{aligned}
 Z &= rX' \\
 E[Z] &= E[rX'] \\
 &= rE[X'] : \text{Using Linearity of expectation} \\
 &= rE[X - 1] : \text{Using the substitution } X' = X - 1 \\
 &= r(E[X] - 1) \\
 &= \frac{r}{p} - R : \text{As } X \sim \text{Geometric}(p), \text{ then } E[X] = 1/p \\
 &= \frac{r(1 - p)}{p}
 \end{aligned}$$

3 Linear Least-Squares Estimation

1. Linear Least Squares Estimate

Linear Least Squares Estimate (LLSE) The LLSE of Y given X , denoted by $L[Y|X]$, is the linear function $a + bX$ that minimizes

$$C(g) = E(|Y - a - bX|^2).$$

Let's try to derive a formula for $L[Y|X]$ in the form of properties of distribution of X and Y

(a) Write $C(g)$ as linear function of $E(Y^2)$, $E(X^2)$, $E(Y)$, $E(X)$ and $E(XY)$

Solution:

$$\begin{aligned} C(g) &= E(|Y - a - bX|^2) \\ &= E(Y^2 + a^2 + b^2X^2 - 2aY - 2bYX + 2abX) \\ &= E(Y^2) + a^2 + b^2E(X^2) - 2aE(Y) - 2bE(YX) + 2abE(X). \end{aligned}$$

(b) Find the values of a and b that minimize the expression in part a. To simplify the calculation use

$$Cov(X, Y) = E(YX) - E(Y)E(X) \text{ and } Var(X) = E(X^2) - E(X)^2.$$

Solution: To find the values of a and b that minimize that expression, we set to zero the partial derivatives with respect to a and b . This gives the following two equations:

$$0 = 2a - 2E(Y) + 2bE(X) \quad (8)$$

$$0 = 2bE(X^2) - 2E(YX) + 2aE(X) \quad (9)$$

From equation 1 we can find a

$$a = E(Y) - bE(X) \quad (10)$$

Substituting that in equation 2 gives

$$0 = 2bE(X^2) - 2E(YX) + 2aE(X) \quad (11)$$

$$= 2bE(X^2) - 2E(YX) + 2[E(Y) - bE(X)]E(X) \quad (12)$$

$$= bE(X^2) - E(YX) + E(Y)E(X) - bE(X)^2 \quad (13)$$

$$= b[E(X^2) - E(X)^2] - [E(YX) - E(Y)E(X)] \quad (14)$$

$$= bVar(X) - Cov(X, Y) \quad (15)$$

$$b = \frac{Cov(X, Y)}{Var(X)} \quad (16)$$

So finally we get values a and b as

$$a = E(Y) - \frac{Cov(X, Y)}{Var(X)}E(X) \quad (17)$$

$$b = \frac{Cov(X, Y)}{Var(X)} \quad (18)$$

(c) Let's put everything together and find the formula for $L[Y|X]$

Solution:

$$\begin{aligned}L[Y|X] &= a + bX \\&= E(Y) - \frac{\text{Cov}(X, Y)}{\text{Var}(X)}E(X) + \frac{\text{Cov}(X, Y)}{\text{Var}(X)}X \\&= E(Y) + \frac{\text{cov}(X, Y)}{\text{var}(X)}(X - E(X))\end{aligned}$$